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Music Genre Classification from Lyrics

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Master's Degree in Artificial Intelligence

Academic Year 2025/2026

Contents

1. Introduction

Music genre classification is a fundamental task in Music Information Retrieval (MIR), with applications ranging from streaming platform recommendation systems to automatic music cataloguing and playlist generation. While most approaches rely on audio features extracted from raw waveforms or spectrograms, lyrics offer a complementary and linguistically rich signal: different genres tend to exhibit characteristic vocabulary, themes, and stylistic patterns. A Hip Hop track typically revolves around urban life and social commentary, whereas Indie Rock lyrics often lean towards introspective and abstract imagery.

This project investigates whether textual features alone are sufficient to distinguish between music genres, comparing a classical machine learning baseline with a modern pre-trained language model. Specifically, we address the following research question:

> *To what extent can song lyrics, processed as natural language, discriminate between music genres, and how much does a contextual language model improve over a bag-of-words baseline?*

We frame the task as a multi-class classification problem over five genres and compare two approaches:

1. **Baseline:** TF-IDF vectorization combined with a Multi-Layer Perceptron (MLP) classifier, representing the classical NLP pipeline.
2. **Advanced:** Fine-tuning of DistilBERT [2], a lightweight distilled variant of BERT [1], which captures contextual and semantic information beyond simple word frequency.

The remainder of this report is organized as follows. Section 2 describes the dataset and preprocessing pipeline. Section 3 details both models. Section 4 presents the experimental setup. Section 5 reports quantitative results and visualizations. Section 6 provides an error analysis. Section 7 concludes with a discussion of findings and possible extensions.

2. Dataset

2.1. Source and Structure

We use the *Scrapped Lyrics from 6 Genres* dataset [4], publicly available on Kaggle. The dataset consists of two CSV files:

- `artists-data.csv`: artist metadata, including a `Genres` field containing one or more semicolon-separated genre labels.
- `lyrics-data.csv`: song lyrics paired with a language code and an artist link used as a foreign key.

The two files are merged on the artist link column to obtain a unified table with columns `lyrics`, `genre`, and `language`.

2.2. Filtering and Genre Selection

The raw merged dataset contains approximately 380,000 entries spanning multiple languages. We apply the following filters:

1. **Language:** retain only English lyrics (`language == "en"`), yielding a monolingual corpus and avoiding confounds introduced by cross-lingual vocabulary differences.
2. **Primary genre:** when multiple genres are listed (semicolon-separated), we keep only the first one, treating it as the primary genre label for that artist.
3. **Top-5 genres:** we retain only songs belonging to the five most frequent genres to ensure sufficient training examples per class and avoid severe class imbalance. After filtering, the dataset contains approximately 75,000 songs distributed across the five selected genres: *Rock*, *Rap*, *Pop*, *Heavy Metal*, and *Indie*.

2.3. Text Preprocessing

Raw lyrics contain structural annotation tags (e.g. `[Chorus]`, `[Verse 1]`), punctuation, numbers, and other non-alphabetic characters that carry no semantic content for genre classification. We apply the following cleaning steps:

1. Remove annotation tags matching the pattern `[\[.*?\]`.
2. Remove all non-alphabetic characters, keeping only ASCII letters and whitespace.
3. Convert to lowercase.
4. Collapse multiple whitespace characters into a single space.

After cleaning, songs with empty text are discarded. The final dataset shape is approximately $(74{,}983 \times 4)$.

2.4. Data Split

The dataset is split into training, validation, and test sets using stratified sampling to preserve class proportions across splits:

Split	Proportion	Approx. samples
Train	70%	52,488
Validation	15%	11,247
Test	15%	11,248

Table 1: Dataset split sizes. Stratified by genre label with `random_state=42`.

3. Methods

3.1. Baseline: TF-IDF + MLP

The baseline pipeline is a classical approach to text classification. It consists of two stages: feature extraction via TF-IDF and classification via a Multi-Layer Perceptron.

3.1.1 TF-IDF Vectorization

Term Frequency–Inverse Document Frequency (TF-IDF) represents each document as a sparse vector of weighted word frequencies. Given a term t in document d from corpus D, the TF-IDF weight is defined as:

$$\mathrm{tfidf}(t, d, D) = \mathrm{tf}(t, d) \cdot \log \frac{|D|}{|\{d' \in D : t \in d'\}|} \tag{1}$$

We use sublinear TF scaling ($\mathrm{tf}(t, d) = 1 + \log(\mathrm{count}(t, d))$) to reduce the impact of very frequent terms. The vocabulary is limited to the top 20,000 terms by frequency, with a minimum document frequency of 2, and includes both unigrams and bigrams (`ngram_range=(1,2)`).

3.1.2 MLP Classifier

The MLP classifier consists of two hidden layers of sizes 512 and 256, with ReLU activations and the Adam optimizer. Early stopping is applied with a patience of 10 iterations on a held-out validation fraction (10% of the training set), preventing overfitting on the sparse TF-IDF features. The output layer uses a softmax activation over the five genre classes.

3.2. Advanced Model: DistilBERT Fine-tuning

3.2.1 Model Architecture

DistilBERT [2] is a distilled version of BERT [1] obtained through knowledge distillation during pre-training. It retains 6 Transformer encoder layers (compared to BERT's 12), reducing the parameter count from 110M to 66M while preserving approximately 97% of BERT's performance on the GLUE benchmark [3].

For sequence classification, a linear classification head is added on top of the `[CLS]` token representation:

$$\hat{y} = \mathrm{softmax}(W \cdot \mathrm{ReLU}(W_0 \cdot h_{\texttt{[CLS]}} + b_0) + b) \tag{2}$$

where $h_{\texttt{[CLS]}} \in \mathbb{R}^{768}$ is the hidden state of the classification token, $W_0 \in \mathbb{R}^{768\times768}$ is the pre-classifier projection, and $W \in \mathbb{R}^{768\times5}$ is the classification head.

3.2.2 Tokenization

We use `DistilBertTokenizerFast` with a maximum sequence length of 256 tokens. Sequences longer than 256 tokens are truncated; shorter ones are padded to the maximum length. The choice of 256 was motivated by the median lyrics length in the dataset, which falls well below this threshold, ensuring minimal information loss from truncation.

3.2.3 Fine-tuning Setup

We fine-tune all model parameters end-to-end using the following configuration:

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	0.01
Batch size	32
Max epochs	4
Warmup steps	10% of total steps
LR schedule	Linear decay with warmup
Gradient clipping	1.0
Mixed precision	fp16 (CUDA only)
Early stopping	Patience = 2 (val. loss)

Table 2: DistilBERT fine-tuning hyperparameters.

The learning rate schedule follows a linear warmup over the first 10% of training steps, followed by linear decay to zero. This is the standard recipe for fine-tuning BERT-family models [1]. Mixed precision training (fp16) is enabled on CUDA devices to approximately halve memory usage and training time.

4. Experimental Setup

4.1. Hardware and Software

Experiments were conducted on Google Colab with an NVIDIA G4 GPU (CUDA 12.8). The software stack includes Python 3.12, PyTorch 2.11, and HuggingFace Transformers 5.10. The TF-IDF baseline uses scikit-learn 1.6.

4.2. Evaluation Protocol

Both models are evaluated on the same held-out test set (15% of the full dataset). We report the following metrics:

- **Accuracy**: fraction of correctly classified samples.
- **F1 macro**: unweighted mean of per-class F1 scores, equally sensitive to all classes regardless of support.
- **F1 weighted**: mean of per-class F1 scores weighted by class support.
- **ROC-AUC (OvR)**: macro-averaged one-vs-rest area under the ROC curve, measuring discriminative ability independently of the classification threshold.

For DistilBERT, the checkpoint with the lowest validation loss is used for final evaluation.

5. Results

5.1. Quantitative Comparison

Table 3 summarizes the performance of both models on the test set. DistilBERT consistently outperforms the TF-IDF + MLP baseline across all metrics, though the margin is modest, approximately 1–2 percentage points in accuracy and F1.

Model	Accuracy	F1 macro	F1 weighted	ROC-AUC OvR
TF-IDF + MLP	0.6421	0.6575	0.6392	0.8839
DistilBERT	0.6562	0.6718	0.6552	0.8945

Table 3: Test set performance of both models. Best values in bold.

The per-class breakdown reveals a consistent pattern across both models (Table 4). **Rap** is by far the easiest genre to classify (F1 ≈ 0.88–0.89), benefiting from a highly distinctive vocabulary. **Indie** is the hardest (F1 ≈ 0.48–0.51), consistently confused with Rock and Pop. **Rock** shows low precision on both models despite high recall, acting as a catch-all class that absorbs misclassifications from neighboring genres.

	TF-IDF + MLP			DistilBERT		
Genre	Prec.	Rec.	F1	Prec.	Rec.	F1
Heavy Metal	0.72	0.76	0.74	0.79	0.69	0.74
Indie	0.53	0.44	0.48	0.61	0.43	0.51
Pop	0.61	0.54	0.57	0.57	0.63	0.60
Rap	0.89	0.88	0.89	0.90	0.86	0.88
Rock	0.58	0.65	0.61	0.58	0.68	0.63

Table 4: Per-class precision, recall and F1 on the test set.

5.2. Training Dynamics

Figure 1 shows the training curves for DistilBERT. The validation loss reaches its minimum at epoch 2 (0.8513), then starts increasing while training loss continues to decrease, a classic sign of overfitting. Early stopping triggers after epoch 4, and the epoch-2 checkpoint is used for final evaluation. The gap between training accuracy (80.2% at epoch 4) and validation accuracy (65.2% at epoch 2) confirms that the model begins to overfit the training distribution from epoch 3 onwards.

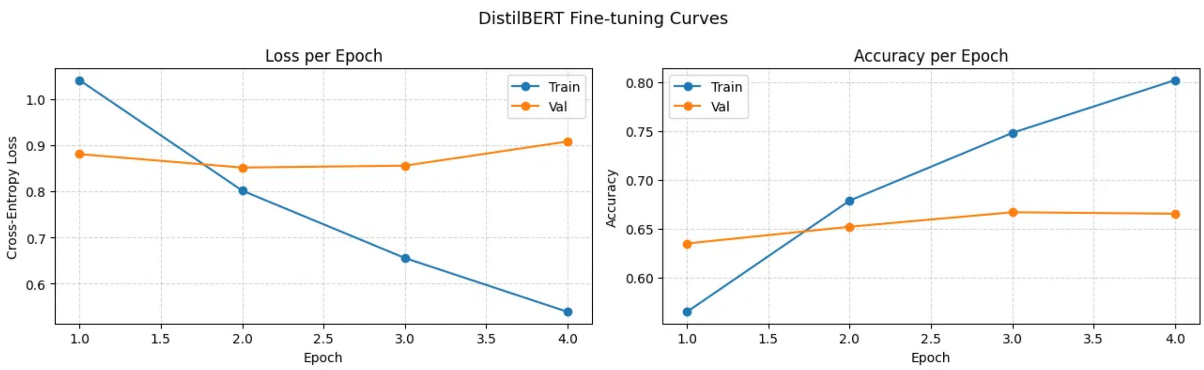

Figure 1: DistilBERT fine-tuning curves. Left: cross-entropy loss. Right: accuracy. The best checkpoint is saved at epoch 2.

5.3. Confusion Matrices

Figure 2 shows the confusion matrices for both models on the test set. Both matrices share a similar structure: Rap and Heavy Metal are well-separated, while Rock, Indie, and Pop form a cluster of mutual confusion. DistilBERT shows a modest improvement in Pop recall (0.54 → 0.63) but slightly lower Heavy Metal recall (0.76 → 0.69), suggesting a different decision boundary rather than a uniformly better model.

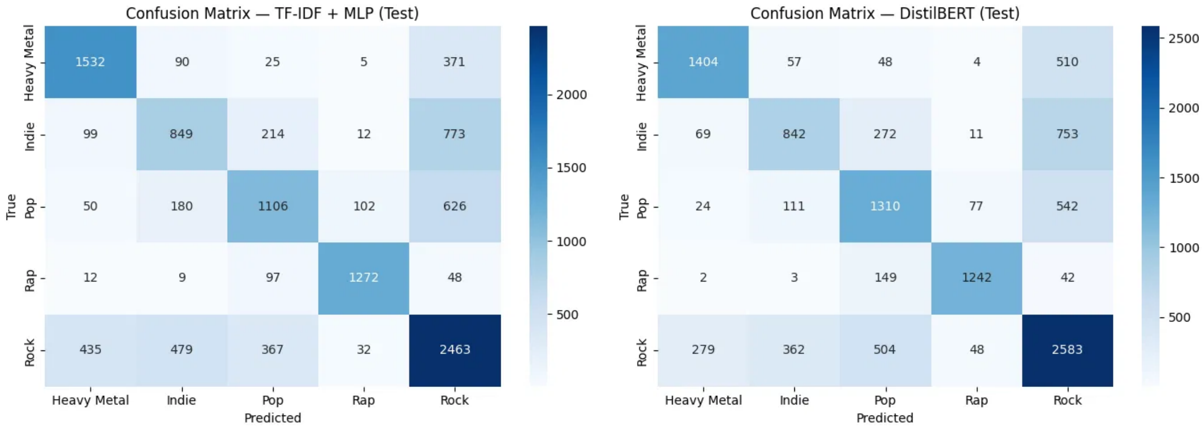

Figure 2: Confusion matrices on the test set. Left: TF-IDF + MLP. Right: DistilBERT.

5.4. ROC Curves

Figure 3 shows the one-vs-rest ROC curves for DistilBERT. The ranking by AUC closely mirrors the per-class F1: Rap achieves near-perfect discrimination (AUC = 0.98), followed by Heavy Metal (0.94), Pop (0.87), Indie (0.86), and Rock (0.81). The relatively low AUC for Rock confirms its role as the most ambiguous genre in the dataset.

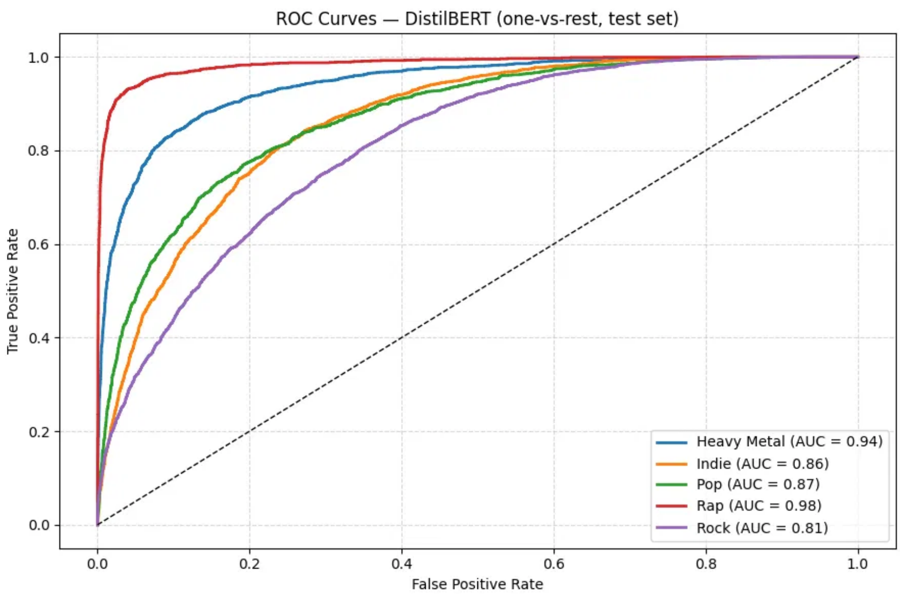

Figure 3: One-vs-rest ROC curves for DistilBERT on the test set.

6. Error Analysis

6.1. Most Confused Genre Pairs

Table 5 lists the most frequent misclassification pairs for DistilBERT on the test set. The dominant error pattern is the confusion between Indie, Pop, Rock, and Heavy Metal — four genres that, at the lyrical level, can overlap significantly. Rap is almost never confused with other genres, confirming its lexical distinctiveness.

True	Predicted	Count
Indie	Rock	753
Pop	Rock	542
Heavy Metal	Rock	510
Rock	Pop	504
Rock	Indie	362
Rock	Heavy Metal	279
Indie	Pop	272
Rap	Pop	149

Table 5: Top misclassification pairs for DistilBERT on the test set.

Rock emerges as the central node of confusion: it is both frequently misclassified as other genres and the most common incorrect prediction for other classes. This is consistent with its status as a broad, stylistically heterogeneous genre whose lyrical content overlaps with Indie (introspective, narrative), Pop (romantic, accessible), and even Heavy Metal (dark imagery).

6.2. Qualitative Analysis on External Lyrics

To further probe the model's behavior, we tested it on seven well-known songs not present in the training dataset. Results are summarized in Table 6.

Song	Artist	True genre	Predicted	Confidence
Shake It Off	Taylor Swift	Pop	Pop	90.9%
Master of Puppets	Metallica	Heavy Metal	Heavy Metal	89.5%
Fate of Ophelia	Taylor Swift	Pop	Heavy Metal	86.9%
Do I Wanna Know?	Arctic Monkeys	Indie	Pop	88.4%
Take Me Out	Franz Ferdinand	Indie	Indie	50.4%
Mr. Brightside	The Killers	Indie Rock	Rock	43.9%
Lose Yourself	Eminem	Rap	Pop	40.5%

Table 6: Qualitative evaluation on external songs not seen during training.

Several observations emerge from this analysis:

- **Tone dominates over artist identity.** The same artist (Taylor Swift) produces opposite predictions depending on lyrical tone: *Shake It Off*, with its light and repetitive vocabulary, is correctly classified as Pop (90.9%), while *Fate of Ophelia*, with its dark literary imagery referencing Shakespeare's Ophelia, is classified as Heavy Metal (86.9%). The model has no knowledge of the artist: it reasons purely on text.

- **Genre-defining vocabulary is key.** *Master of Puppets* (Metallica) is classified correctly with high confidence because its vocabulary is strongly associated with Heavy Metal in the training data. *Lose Yourself* (Eminem), despite being a canonical Rap track, is classified as Pop, likely because its narrative, motivational tone does not match the more typical Rap vocabulary in the training corpus.

- **Indie is the hardest genre to pin down.** Of three Indie songs tested, only one (*Take Me Out*) is correctly classified, and with only 50.4% confidence. *Do I Wanna Know?* is classified as Pop due to its romantic and introspective language, and *Mr. Brightside* is classified as Rock with Pop close behind. This mirrors the quantitative results, where Indie has the lowest F1 of all five genres.

- **Misclassification confidence can be misleadingly high.** Fate of Ophelia and Do I Wanna Know? are both misclassified with confidence above 85%, suggesting that the model is not well-calibrated in cases where lyrical tone conflicts with musical genre.

These findings highlight a fundamental limitation of lyrics-only classification: musical genre is a multidimensional concept determined by instrumentation, rhythm, production style, and cultural context, none of which are captured by text alone.

7. Conclusions

This project investigated music genre classification from lyrics, comparing a TF-IDF + MLP baseline with a fine-tuned DistilBERT model on a dataset of approximately 75,000 English songs across five genres.

The main findings can be summarized as follows:

- **DistilBERT outperforms the baseline**, but only marginally, by approximately 1–2 percentage points across all metrics. This suggests that for this task, the additional representational power of a contextual language model yields diminishing returns compared to a well-tuned bag-of-words approach.

- **Genre difficulty is highly unequal.** Rap is classified with F1 $\approx$ 0.88 on both models, while Indie achieves only F1 $\approx$ 0.48–0.51. The difficulty correlates with lexical distinctiveness: genres with a characteristic vocabulary (Rap, Heavy Metal) are easy; genres with overlapping lyrical styles (Indie, Rock, Pop) are hard.

- **The model reasons on tone and vocabulary, not musical identity.** The qualitative analysis on external songs reveals that the classifier associates dark/dramatic language with Heavy Metal, romantic/simple language with Pop, and narrative language with Pop or Indie, regardless of the actual artist or genre. This is an expected but important limitation of text-only approaches.

- **Rock acts as a catch-all class**, absorbing misclassifications from all other genres. Its broad stylistic range makes it the most ambiguous label in the dataset.

Limitations. The most significant limitation is the text-only nature of the approach. Genre is ultimately a musical concept: two songs with identical lyrics but different instrumentation can belong to entirely different genres. A multimodal approach combining lyrics with audio features (e.g. MFCCs, mel spectrograms) would likely yield substantially better results. Additionally, the genre labels in the dataset are artist-level rather than song-level, introducing label noise for artists who span multiple genres.

Future work. Several extensions could improve on these results: (1) multimodal fusion of lyrics and audio features; (2) data augmentation to address class imbalance and the low Indie recall; (3) interpretability analysis using attention weights or SHAP values to identify which lyrical patterns drive each genre prediction; (4) experimenting with larger language models such as RoBERTa or domain-adapted models trained on song lyrics.

References

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," in *Proc. NeurIPS EMC^2 Workshop*, 2019.

[3] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman, "GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding," in *Proc. ICLR*, 2019.

[4] Neisse, "Scrapped Lyrics from 6 Genres," Kaggle, 2020. https://www.kaggle.com/datasets/neisse/scrapped-lyrics-from-6-genres